POTABLE WATER TREATMENT & PRESSURIZED DISTRIBUTION SCHEMATIC:
1. Raw water storage capacity: 500,000 Litres underground RCC reservoir with food-grade epoxy coating.
2. Central filtration train: 2x 75 m3/hr Multi-Grade Sand Filters (MGF) followed by 2x Activated Carbon Filters (ACF).
3. Drinking water Reverse Osmosis (RO) system: 50 KLD capacity with 2:1 staging, cartridge pre-filters, and UV disinfection.
4. Hydro-Pneumatic Pumping Skid: 3x vertical multistage centrifugal pumps (2 Working + 1 Standby) operating on constant pressure control.
5. Pre-charged nitrogen diaphragm expansion tank (500L) dampening hydraulic surges and cycling pulses across terminal fixtures.

| Equipment Tag | Equipment Description | Capacity | Operating Pressure | Motor Rating | MOC |
|---|---|---|---|---|---|
| MGF-01/02 | Multi-Grade Sand Filters | 75 m3/hr each | 3.5 bar | N/A (Pressure Vessel) | FRP / MS Epoxy Lined |
| ACF-01/02 | Activated Carbon Filters | 75 m3/hr each | 3.5 bar | N/A (Pressure Vessel) | FRP / MS Epoxy Lined |
| RO-SKID | Reverse Osmosis Membrane Unit | 50 KLD | 14.0 bar | 11.0 kW HP Pump | SS 316 High Pressure |
| HYPN-PMP-01 | Booster Pump 1 (Duty) | 45 m3/hr | 4.5 bar | 7.5 kW Vertical Multi | SS 304 Stainless Steel |
| HYPN-PMP-02 | Booster Pump 2 (Duty) | 45 m3/hr | 4.5 bar | 7.5 kW Vertical Multi | SS 304 Stainless Steel |
| HYPN-PMP-03 | Booster Pump 3 (Standby) | 45 m3/hr | 4.5 bar | 7.5 kW Vertical Multi | SS 304 Stainless Steel |

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| AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY | | |
| DWG TITLE: WATER SUPPLY TREATMENT PLANT & HYDRO-PNEUMATI | | |
| DRAWING NO: SL-288-WS-99 | REV: R1 | SCALE: 1:100 |
| CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH | DATED: 28-09-2026 | |